جامعة أم القرى
UMM AL-QURA UNIVERSITY

PROJECT TITLE:

Heart Disease Prediction Using Machine Learning

ABSTRACT

This project aims to predict heart disease in patients using supervised machine learning. The dataset consists of 918 patients and 11 clinical features obtained from Kaggle. Two classification models were applied: Logistic Regression and Random Forest. Random Forest achieved the best performance with an accuracy of 0.89, outperforming Logistic Regression at 0.85.

INTRODUCTION

Heart disease is one of the leading causes of death globally. Early and accurate diagnosis is critical for effective treatment. This project addresses the problem of predicting heart disease using machine learning techniques. The main objective is to build and compare classification models that can identify whether a patient has heart disease based on clinical measurements. The research question is: *Can we accurately classify patients as having heart disease or not using supervised learning models?* This is a binary classification problem, making supervised learning approaches suitable.

METHODOLGY

Data Source & Description:

- Source: Kaggle — https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction
- 918 rows × 12 columns
- 7 numeric columns, 5 categorical columns
- Target: HeartDisease (1 = Disease, 0 = Normal)

Data Preprocessing:

- Checked for missing values → No missing values found

- Used LabelEncoder to convert categorical columns (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope)
- Split the data: 80% train and 20% test
- Applied StandardScaler to normalize numeric features

Algorithm/Model Selection:

- **Logistic Regression:** Selected because it is simple and suitable for binary classification problems
- **Random Forest:** Selected because it is more powerful and provides feature importance
- Tested multiple values of C for Logistic Regression and selected C=1
- Tested multiple values of n_estimators and selected 50

Training Procedure:

- Train = 80%
- Test = 20%

- Random State = 42

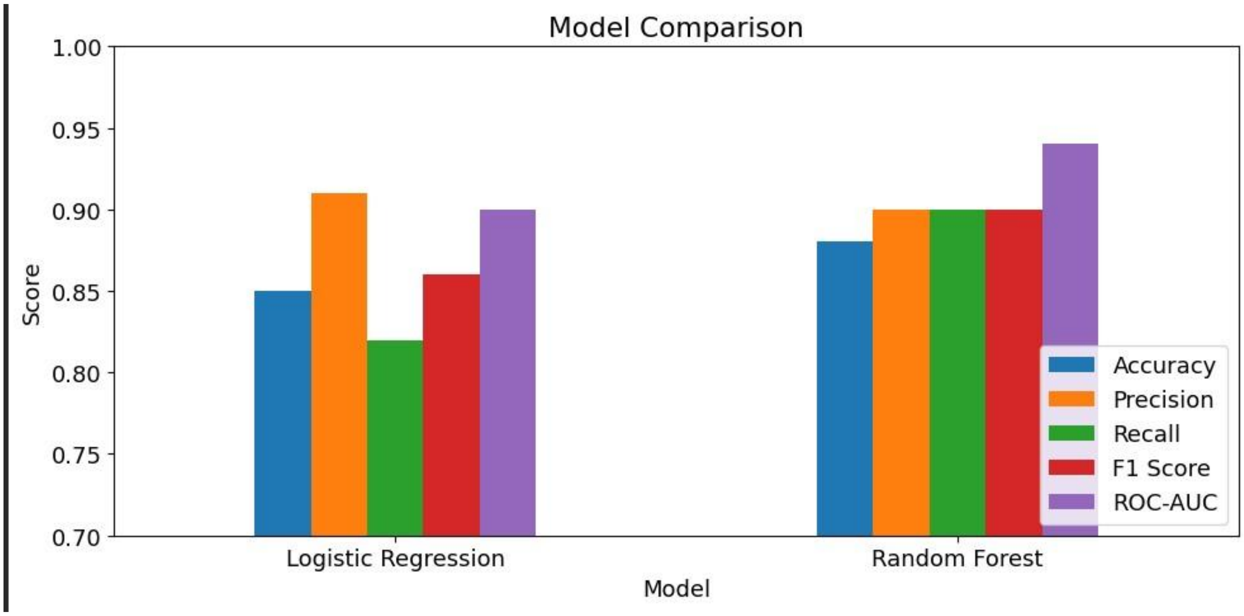
Evaluation Strategy:

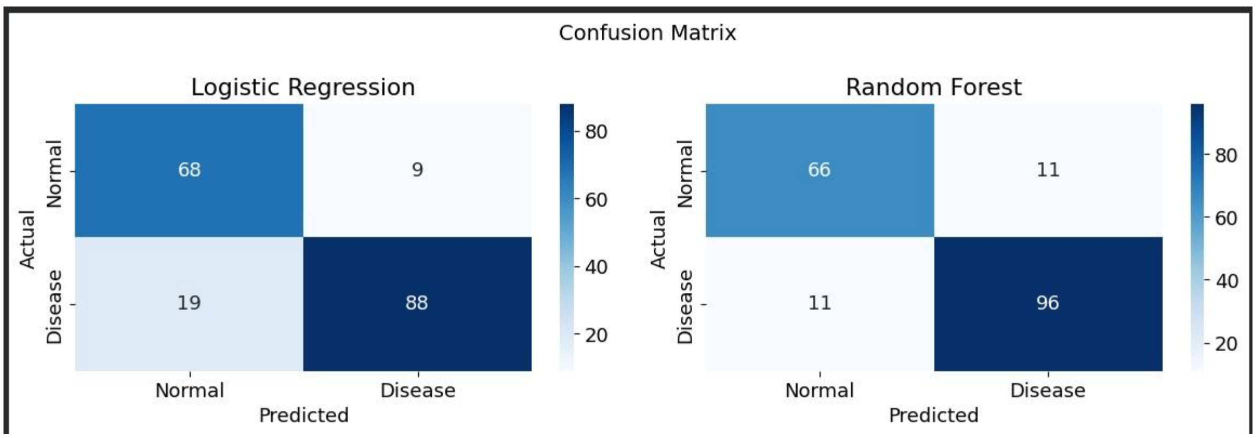
- Metrics used: Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Recall was prioritized because the project is medical — predicting a sick patient as healthy is the most dangerous error

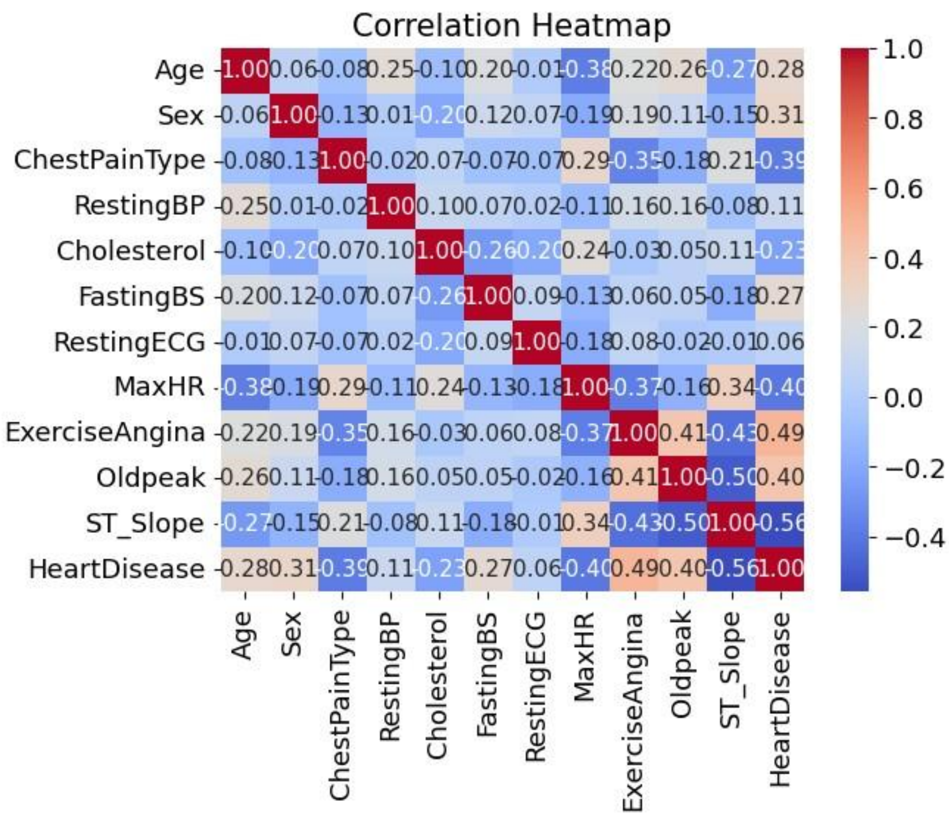
Libraries and functions:

- pandas, scikit-learn, matplotlib, seaborn

RESULTS AND DISCUSSION







	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Model					
Logistic Regression	0.85	0.91	0.82	0.86	0.90
Random Forest	0.88	0.90	0.90	0.90	0.94

Interpretation:

Random Forest outperformed Logistic Regression across all metrics, achieving an accuracy of 0.88 vs 0.85, Recall of 0.90 vs 0.82, and ROC-AUC of 0.94 vs 0.90.

The Confusion Matrix shows that Random Forest produced only 11 false negatives compared to 19 in Logistic Regression, meaning fewer sick patients were misclassified as healthy. This makes Random Forest a safer model in a medical context where missing a sick patient is the most critical error.

Both models were trained on 80% of the dataset and evaluated on the remaining 20% . The test results are consistent with the training performance, suggesting that both models generalize well to unseen data without signs of overfitting.

The Correlation Heatmap reveals that ST_Slope (0.56), ExerciseAngina (0.49), and Oldpeak (0.40) are the strongest predictors of heart disease, while MaxHR (-0.40) indicates that lower maximum heart rates are associated with higher disease risk, consistent with known clinical risk factors.

CONCLUSION

This project aimed to predict whether a patient has heart disease based on clinical features using supervised machine learning. Two models were applied: Logistic Regression and Random Forest. Random Forest achieved the best performance with an accuracy of 0.88, a Recall of 0.90, and a ROC-AUC of 0.94, outperforming Logistic Regression across all metrics. These results suggest that machine learning models can serve as a useful tool to assist medical professionals in early heart disease detection. One of the main challenges encountered was the presence of zero values in the Cholesterol column, which may represent missing or invalid data and could affect model performance. Future work could explore more advanced models such as XGBoost or Neural Networks, apply k-fold crossvalidation for more robust evaluation, or collect larger datasets to improve generalization.